

Biomedical Signal Processing: ECG Noise Filtering

Department of Science and Technology, Humber College North Campus

MENG-3520: Signal Processing

Prepared For: Dr. Saba Sedghizadeh

Prepared By: Hamza Hersi, Keith Lingat

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Introduction

Clinical Background and Motivation

An Electrocardiogram (ECG) is a critical, non-invasive diagnostic tool used to passively record the electrical activity of the heart. By placing small electrode patches on a patient's body, medical professionals can capture the electrical signatures of individual heartbeats, mathematically represented as the P-QRS-T complex. The most prominent feature of this waveform is the QRS complex, which signifies the depolarization and contraction of the heart's main ventricular chambers. Accurate analysis of these specific waveforms is essential for diagnosing severe cardiac arrhythmias, such as tachycardia (abnormally fast heart rate) and bradycardia (abnormally slow heart rate).

The Signal Processing Problem

Despite its clinical importance, raw ECG data is highly susceptible to environmental and physiological noise, which can severely compromise the integrity of the signal. In this project, two primary sources of interference are addressed:

- **Baseline Wander:** A low-frequency signal drift primarily caused by patient respiration and physical movement. This causes the entire ECG waveform to slowly roll up and down the amplitude axis.
- **Power-Line Interference:** A high-frequency, 60 Hz electromagnetic hum introduced by the local electrical grid and the monitoring equipment itself.

If left unmitigated, these artifacts can obscure vital clinical features, leading to false arrhythmia detections or inaccurate heart rate calculations.

Literature Review and Frequency Characteristics

Standard biomedical literature establishes that the most critical frequency components of a healthy human ECG signal typically reside between 0.5 Hz and 40 Hz. Because baseline wander generally exists below 0.5 Hz, and power-line interference exists precisely at 60 Hz, digital signal processing (DSP) techniques can be employed to target and remove these specific noise bands. Filtering must be done carefully to attenuate the noise without causing phase distortion or amplitude attenuation to the true P-QRS-T complex.

Project Objectives

The primary objective of this project is to design and simulate a robust digital signal processing pipeline capable of denoising a raw ECG signal. By systematically applying discrete-time filters, the system aims to flatten the baseline drift and eliminate electrical interference. Ultimately, this clean signal will be processed through a peak detection algorithm to accurately extract and calculate the patient's heart rate in various physiological states.

Signal Construction & System Modeling

Signal Acquisition and Characteristics

To effectively design and test the digital signal processing pipeline, a reliable signal source was required. For this project, a Synthetic ECG Dataset (RRSYNTHdata.mat) was utilized to simulate a realistic clinical environment. The continuous-time physiological signal was converted into a discrete-time digital format using a sampling rate (f_s) of 250 Hz. According to the Nyquist-Shannon sampling theorem, the sampling rate must be at least twice the highest frequency present in the signal to prevent aliasing. Given that the primary frequency components of a standard ECG signal reside between 0.5 Hz and 40 Hz, the selected sampling rate of 250 Hz establishes a Nyquist frequency of 125 Hz. This safely exceeds the minimum requirement, ensuring that the critical P-QRS-T waveform features are preserved in the digital domain without distortion.

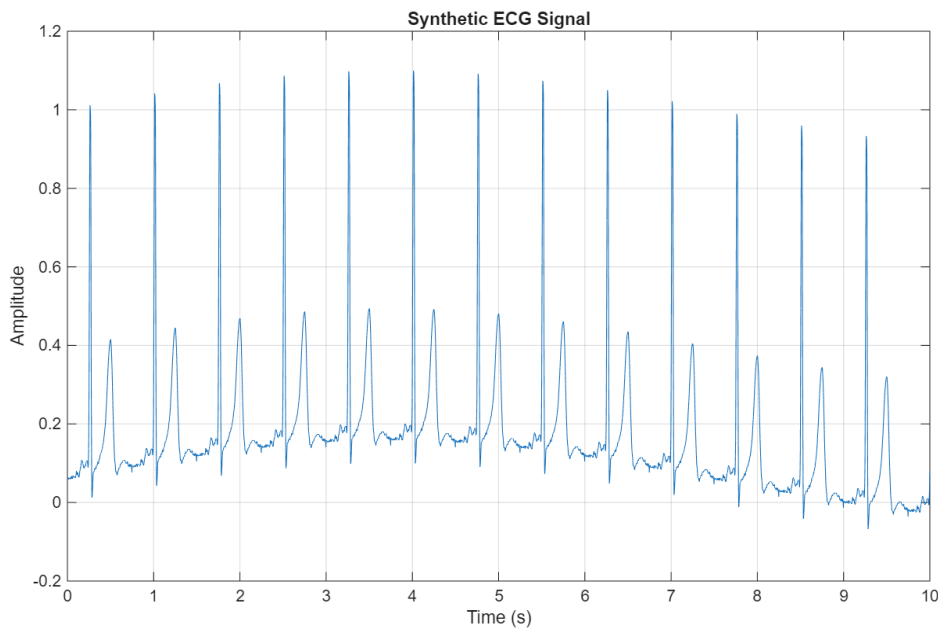


Figure 1: 10-second window of the raw, discrete-time synthetic ECG signal

Mathematical System Modeling

Before developing the filtering algorithms, it is necessary to mathematically model the noisy input signal. The raw, discrete-time ECG signal can be represented as a linear combination of the desired true signal and the additive noise components. The system is modeled using the following equation:

$$ECG(n) = HB(n) + BW(n) + PI(n)$$

Where:

- $ECG(n)$ represents the raw, unrecorded discrete-time ECG signal captured by the sensor system.
- $HB(n)$ represents the true, clean heartbeat signal containing the vital P-QRS-T complexes.
- $BW(n)$ represents the low-frequency baseline wander artifact, typically introduced by patient respiration and movement.
- $PI(n)$ represents the high-frequency power-line interference, specifically the 60 Hz electromagnetic hum from the local power grid.

The core engineering objective of the subsequent filter design is to isolate $s[n]$ by mathematically attenuating the $b[n]$ and $i[n]$ components from the raw input $x[n]$.

Time-Domain & Frequency-Domain Analysis

Time-Domain Analysis

Time-domain analysis evaluates how a signal's amplitude changes over a specific continuous period, allowing for the direct observation of waveform shapes and overall system responses. When analyzing the raw, unfiltered synthetic ECG signal in the time domain, the most prominent abnormality is a massive, drifting offset. This offset is the physical manifestation of baseline wander, which causes the entire P-QRS-T complex to shift away from the zero-amplitude baseline. By plotting the raw signal against the final filtered output, the effectiveness of the digital processing pipeline becomes visually evident. The filtered signal successfully remains centered at zero amplitude, demonstrating that the low-frequency drift has been eliminated without altering the critical timing or shape of the individual R-peaks.

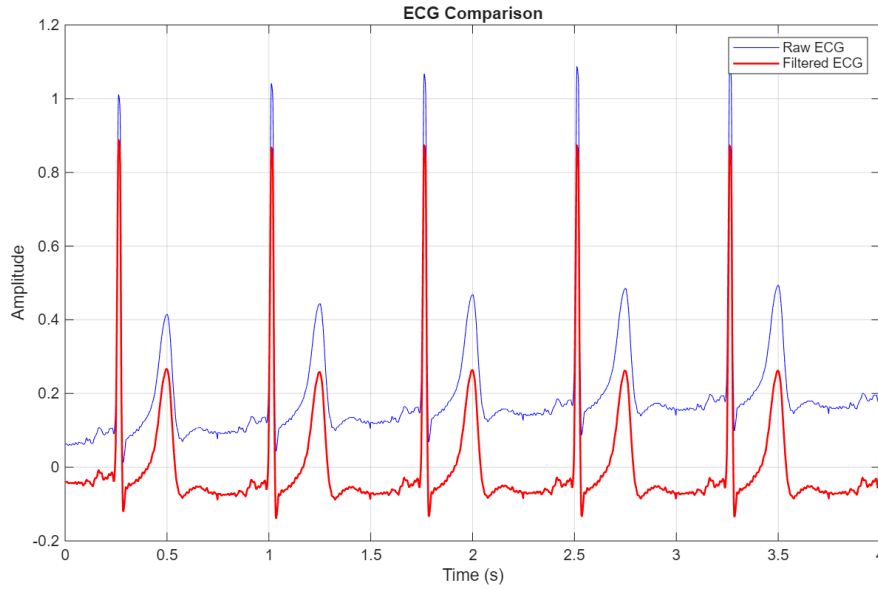


Figure 2: Time-domain comparison demonstrating Raw ECG signal vs Filtered ECG signal

Frequency-Domain Analysis

While the time domain illustrates when events happen, the frequency domain reveals the specific spectral content comprising the signal. Applying a Fast Fourier Transform (FFT) to the discrete-time data converts it into a magnitude spectrum, making it possible to identify the exact frequencies of the noise components.

In the pre-filtered magnitude spectrum, the mathematical representation of the baseline wander is clearly visible as an overwhelming amplitude spike centered exactly at 0 Hz.

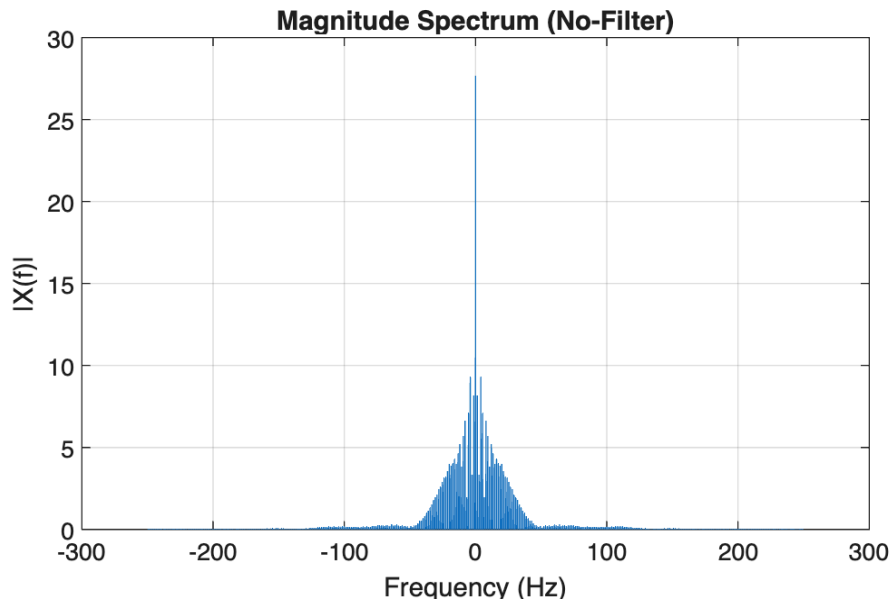


Figure 3: Frequency-domain representation of raw ECG signal

After passing the signal through the processing pipeline, the frequency-domain analysis confirms the success of the filter designs. The spectrum reveals that the low-frequency DC offset is completely attenuated, leaving the critical 0.5 Hz to 40 Hz physiological band intact. Additionally, the spectral energy at the 60 Hz mark is extracted, verifying that the power-line interference has been successfully neutralized while preserving the vital heartbeat frequencies.

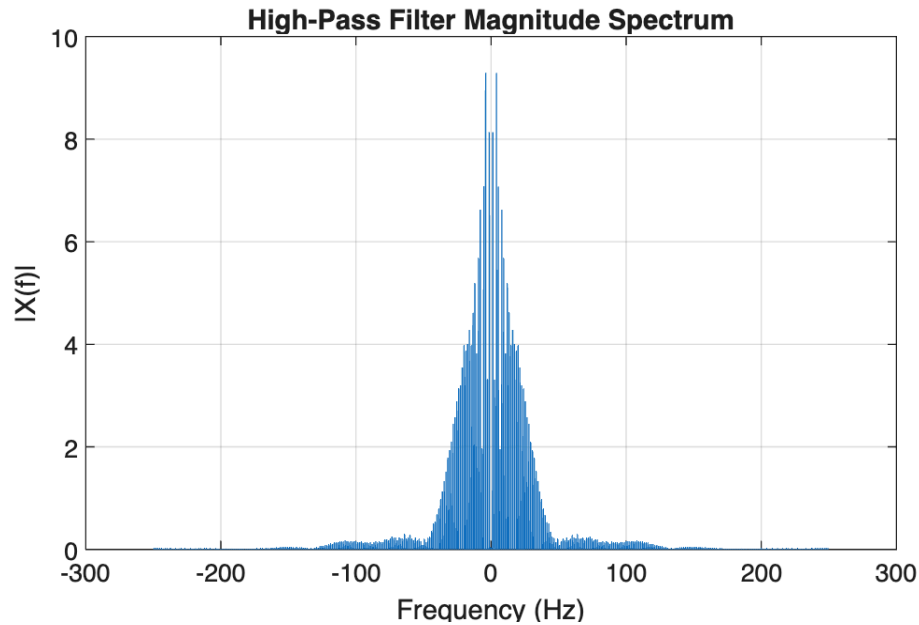


Figure 4: Magnitude spectrum after 4th-order Butterworth filter

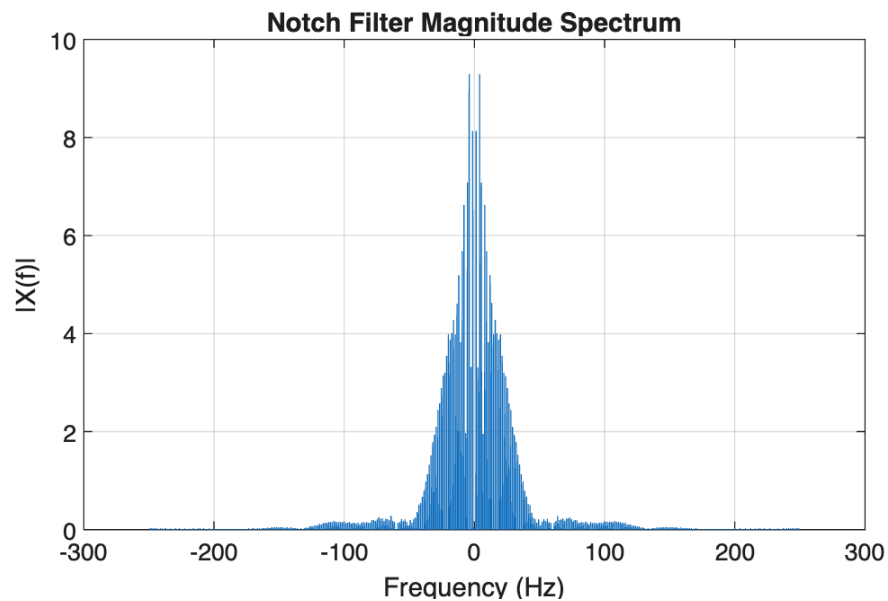


Figure 5: Final magnitude spectrum demonstrating combined effects of high-pass and notch filters

Design Methodology

Systematic Filtering Approach

To successfully extract the true ECG signal ($s[n]$) from the noisy acquisition, a dual-stage digital signal processing pipeline was designed. This approach systematically targets the low-frequency baseline wander and the high-frequency power-line interference using two distinct linear time-invariant (LTI) filters.

Stage 1: High-Pass Filter Design

To eliminate the baseline wander, a 4th-order Butterworth high-pass filter was designed. The Butterworth topology was specifically selected for its maximally flat passband response. This characteristic is critical in biomedical signal processing, as it ensures no artificial ripples or phase distortions are introduced into the physiological P and T waves of the heartbeat.

The filter was configured with a precise cutoff frequency (f_c) of 0.5 Hz to specifically isolate and attenuate the slow physical drift. The filter was initially modeled in the continuous-time Laplace domain (s-domain) and subsequently mapped to the discrete-time Z-domain using the bilinear (Tustin) approximation method. Analyzing the pole-zero map confirms that all poles are located strictly inside the unit circle, ensuring absolute system stability.

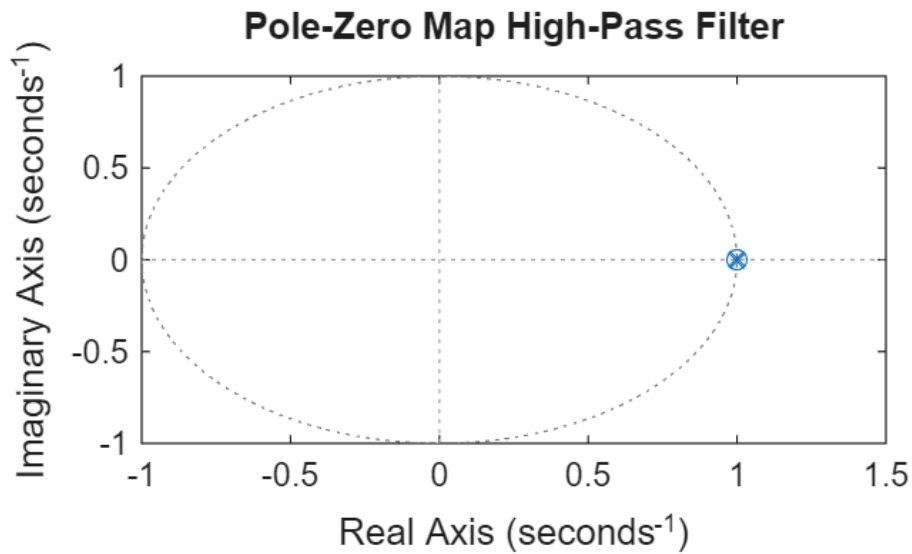


Figure 6: Pole-zero map of the 4th-order Butterworth high-pass filter

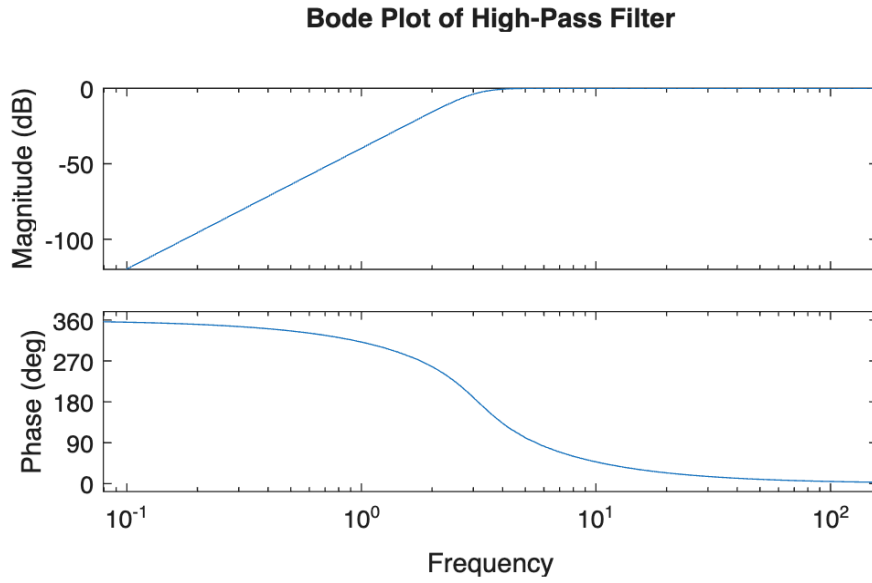


Figure 7: Bode plot of the high-pass filter

Stage 2: Power-Line Interference Mitigation

Following the removal of the baseline drift, the signal required mitigation of the 60 Hz electromagnetic grid hum. To address this, a 2nd-order Infinite Impulse Response (IIR) Bandstop (Notch) filter was implemented. Using MATLAB's `designfilt` function, the filter was engineered to target the exact half-power frequencies between 59 Hz and 61 Hz. This acts as a frequency cutoff, neutralizing the electrical buzz without degrading the high-frequency components of the QRS complex.

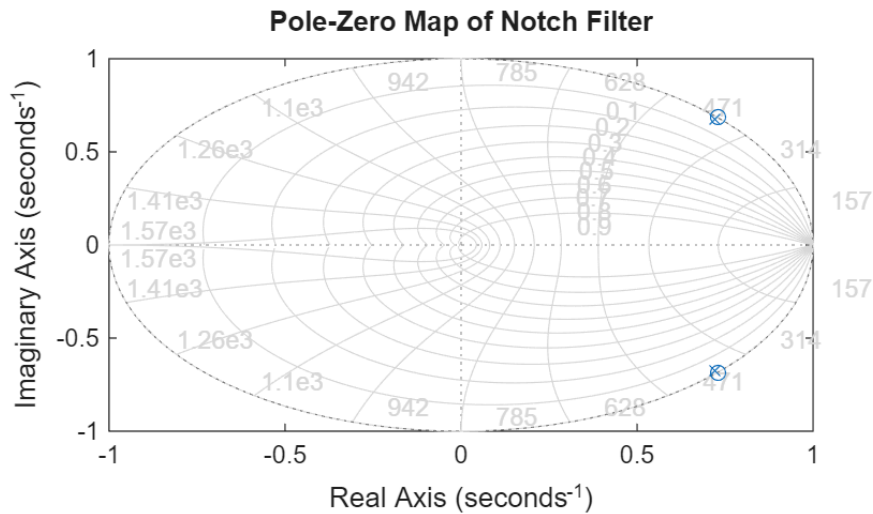


Figure 8: Pole-zero map of the 2nd-order notch filter

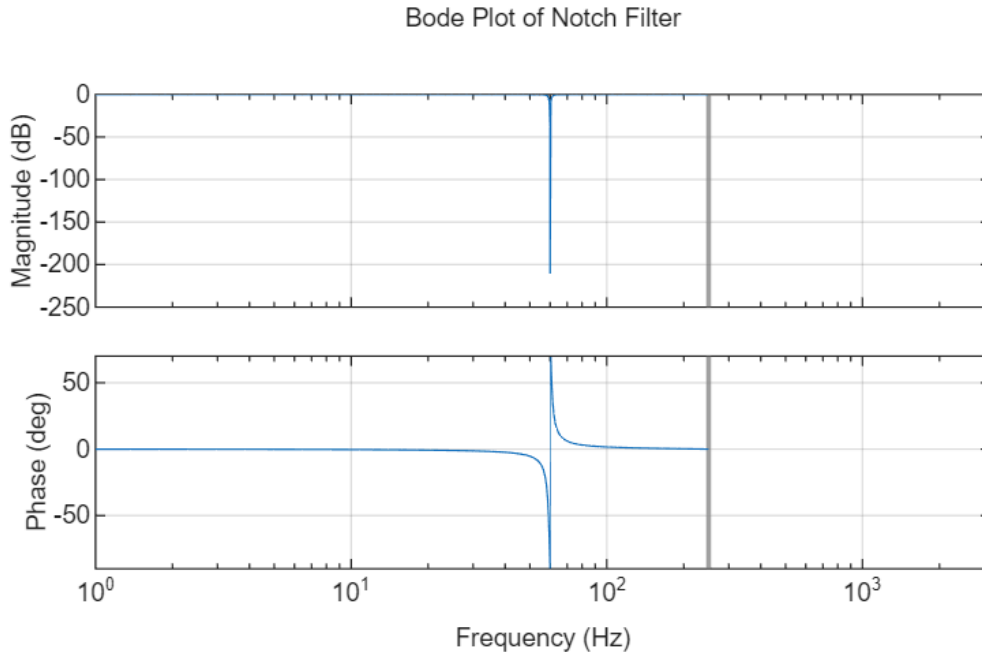


Figure 9: Bode plot of the notch filter

Algorithm Implementation and Zero-Phase Distortion

A critical consideration in ECG analysis is the preservation of time-domain features; if the filter shifts the signal in time, the calculated heart rate will be fundamentally inaccurate. Standard digital filters introduce phase delays. To circumvent this, the filtering pipeline was implemented using MATLAB's `filtfilt` function. This algorithm performs zero-phase digital filtering by processing the input data in both the forward and reverse directions. The result is a mathematically clean signal with precisely zero phase distortion, ensuring the R-peaks remain in their exact, original temporal locations for the detection phase.

Testing & Performance Assessment

Peak Detection Algorithm

To evaluate the effectiveness of the digital filtering pipeline, a performance assessment was conducted by extracting the heart rate from the denoised ECG signal. This was achieved using MATLAB's `findpeaks` algorithm to locate the precise temporal coordinates of each R-peak within the QRS complexes.

To ensure the algorithm accurately isolated the R-peaks without registering false positives, specific detection thresholds were established:

1. **Minimum Peak Height:** Set to 40 percent of the signal's maximum amplitude. This threshold safely ignores the smaller physiological P and T waves, as well as any residual high-frequency artifacts.
2. **Minimum Peak Distance:** Set to 0.6 seconds. This prevents the algorithm from double-counting wide, abnormal waveforms or multi-phasic complexes.

Heart Rate Calculation

Once the R-peaks were successfully indexed, the system calculated the time difference (Δt) between consecutive peaks (the R-R interval). The instantaneous heart rate in beats per minute (bpm) was derived using the following mathematical relationship:

$$HR = \frac{60}{\Delta t_{RR}}$$

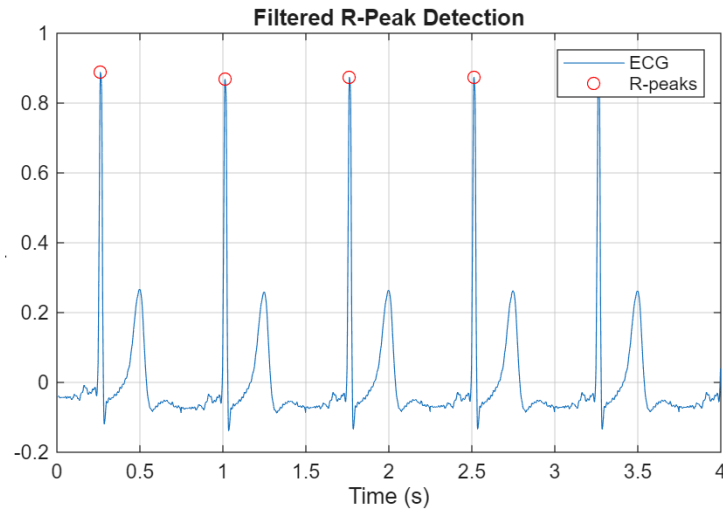


Figure 10: Successful identification of R-peaks

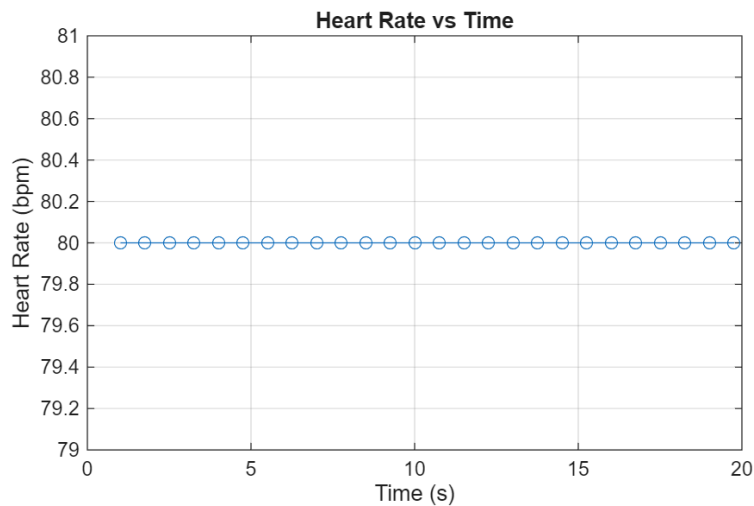


Figure 11: Calculated 80 bpm heart rate plotted over time

System Robustness and Other Cases

To further validate the robustness of the detection system, it was tested against simulated pathological conditions, specifically bradycardia (abnormally slow heart rate) and tachycardia (abnormally fast heart rate).

When fed a simulated bradycardia dataset, the system successfully identified the spaced-out R-peaks and tracked a steady heart rate of 60 bpm. Conversely, when tested against a simulated tachycardia dataset, the algorithm proved it could handle a higher frequency of complexes, accurately tracking a steady rate of 90 bpm without the 0.6-second minimum distance threshold causing interference.

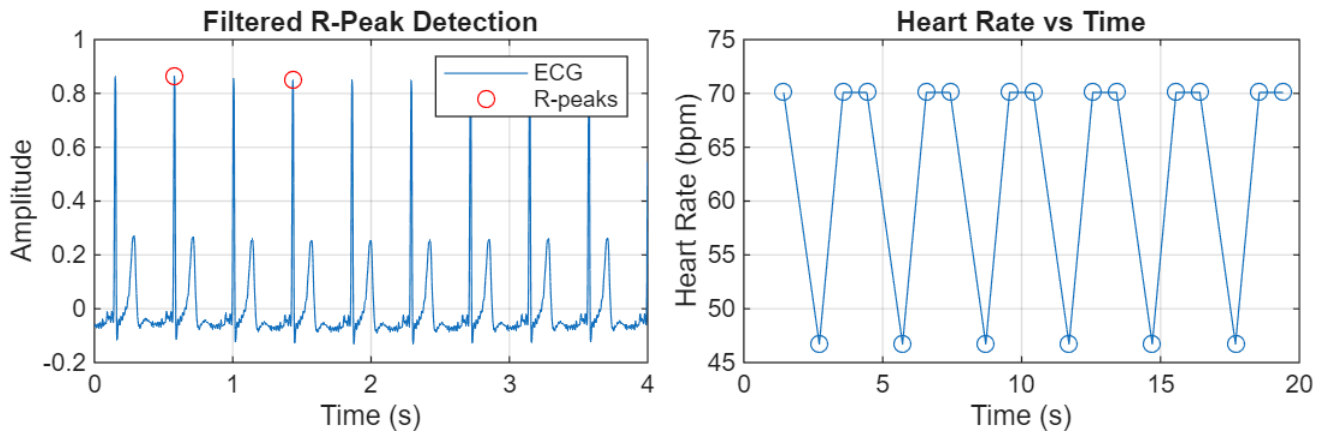


Figure 12: System performance evaluation on a bradycardia signal

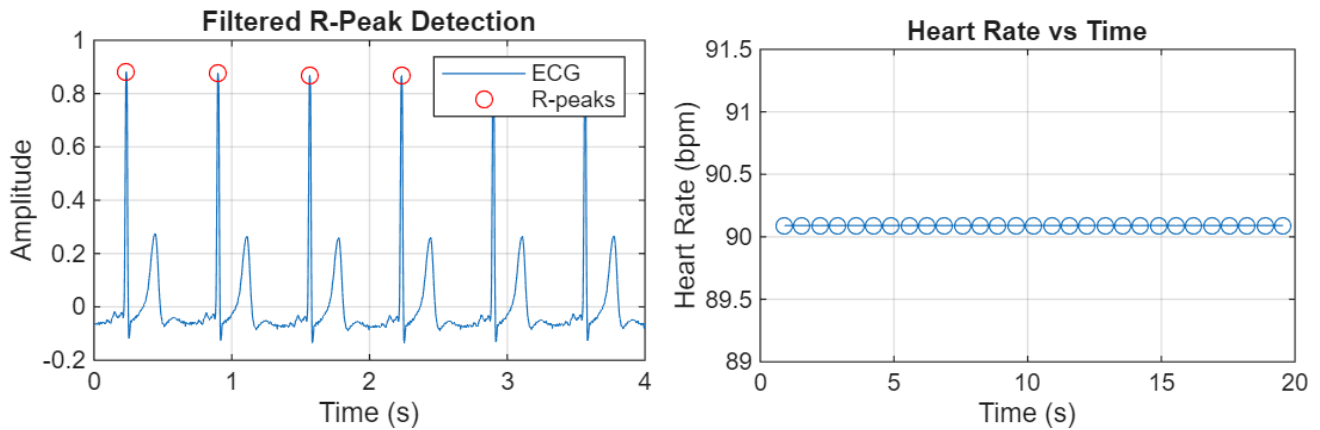


Figure 13: System performance evaluation on a tachycardia signal

Conclusion, Discussion & Future Work

Conclusion

The primary objective of this project was to design a robust digital signal processing pipeline capable of extracting accurate medical metrics from a noisy Electrocardiogram (ECG) signal. This objective was successfully met through the implementation of a dual-stage filtering system. As demonstrated in the time-domain and frequency-domain plots, the 4th-order Butterworth high-pass filter successfully flattened the low-frequency baseline wander, while the 2nd-order Notch filter precisely removed the 60 Hz power-line interference. By utilizing zero-phase filtering `filtfilt`, the critical P-QRS-T clinical features were preserved without temporal distortion. Consequently, the peak detection algorithm accurately and reliably extracted heart rates across resting (80 bpm), bradycardia (60 bpm), and tachycardia (90 bpm) simulated conditions.

Discussion of Limitations

While the simulated system performed with high accuracy, there are notable engineering limitations in the current design. First, the reliance on MATLAB's `filtfilt` function inherently makes the system non-causal. Because this algorithm processes the data in both the forward and backward directions to prevent phase shifting, it requires the entire dataset to be pre-recorded. Therefore, this specific code cannot be deployed for live patient monitoring. Furthermore, the peak detection algorithm relies on static, hardcoded parameters (a 40 percent height threshold and a 0.6-second minimum distance). While sufficient for the controlled test cases, a fixed 0.6-second delay mathematically limits the system from detecting heart rates exceeding 100 bpm. In a real-world clinical scenario, a patient experiencing severe tachycardia could have their peaks erroneously ignored by this static threshold.

Future Work

To bridge the gap between this theoretical simulation and a viable biomedical product, future iterations of this project must address real-time processing capabilities. Upgrading the software architecture to utilize causal IIR filtering for live monitoring rather than relying on non-causal, post-processing functions like `filtfilt` would allow the pipeline to process data sequentially and run directly on a wearable device. Additionally, using an Adaptive Filter (Least Mean Squares algorithm) could be introduced to dynamically handle unpredictable, active muscle noise (EMG interference) that occurs when a patient physically moves or exercises during monitoring.

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